

KHUZIR, Russian FederationWMO: **306370**Lat: **53.190N**Lon: **107.330E**Elev: **1514**StdP: **13.91**Time Zone: **8.00 (E08)**Period: **86-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-24.3	-19.3	-29.3	1.1	-23.0	-25.0	1.5	-18.4	23.3	6.2	21.5	9.6	5.0	260	0.685

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	13.1	72.5	61.8	70.0	60.3	67.9	59.1	63.5	69.9	61.8	67.9	60.4	66.3	5.8	270

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
60.8	84.1	66.5	59.1	79.2	65.5	57.5	74.6	64.6	29.7	70.5	28.5	67.9	27.4	66.4	70.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 23.4	21.4	17.9	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(5)			-27.3	66.6	5.2	2.2	-31.1	68.2	-34.2	69.5	-37.1	70.7	-40.9	72.2		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	31.7	0.2	-0.3	15.4	31.5	42.0	51.8	60.0	60.0	48.2	34.0	20.8	14.1		
	DBStd	21.99	10.41	10.76	11.53	7.27	6.01	5.09	4.50	4.84	6.48	6.48	8.81	8.12		
	HDD50	7470	1543	1408	1072	554	257	38	1	3	110	496	876	1112		
	HDD65	12185	2008	1828	1537	1004	713	397	161	168	505	961	1326	1577		
	CDD50	779	0	0	0	0	9	91	312	312	55	0	0	0		
	CDD65	19	0	0	0	0	0	0	7	12	0	0	0	0		
	CDH74	33	0	0	0	0	0	0	10	22	1	0	0	0		
	CDH80	1	0	0	0	0	0	0	0	1	0	0	0	0		
	WSAvg	6.7	6.9	5.7	6.1	6.6	6.5	5.0	5.0	5.3	7.0	7.9	9.6	9.0		
	PrecAvg	14.0	0.5	0.3	0.3	0.6	1.1	2.0	2.9	2.9	1.5	0.6	0.7	0.6		
(15) Precipitation	PrecMax	17.2	1.9	0.6	0.9	1.2	2.0	4.3	5.8	4.8	3.3	1.5	1.2	1.8		
	PrecMin	11.0	0.1	0.0	0.0	0.2	0.3	0.8	1.5	1.3	0.6	0.1	0.3	0.2		
	PrecStd	1.7	0.3	0.1	0.2	0.2	0.4	0.8	1.0	1.1	0.7	0.3	0.2	0.3		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	26.5	33.9	44.2	56.9	65.7	68.6	75.0	76.4	68.0	54.0	44.9	37.9		
		MCWB	22.5	28.3	36.1	44.7	49.1	56.2	61.1	63.2	58.3	44.5	37.4	32.5		
	2%	DB	22.0	27.3	39.4	50.3	59.4	65.6	72.3	73.2	64.2	50.1	40.6	31.2		
		MCWB	18.9	23.4	32.6	40.0	46.5	54.3	61.2	63.0	55.6	42.8	33.7	26.7		
	5%	DB	18.8	21.3	36.0	46.0	55.2	63.2	70.2	70.5	61.6	47.6	37.0	28.2		
		MCWB	16.5	18.1	30.1	37.3	45.2	53.1	60.1	61.3	55.0	41.4	31.3	24.5		
	10%	DB	15.7	16.2	33.1	42.4	51.9	60.7	68.2	68.3	59.1	44.9	33.4	26.0		
		MCWB	13.5	13.7	28.1	35.2	42.9	52.0	59.1	60.1	53.1	38.8	28.8	22.5		
	0.4%	WB	23.1	28.7	36.0	44.9	51.5	59.4	64.7	66.5	59.7	46.4	38.2	32.4		
		MCDB	26.5	33.7	42.3	55.0	61.7	65.4	70.7	72.5	64.9	51.3	44.4	38.0		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	19.2	23.4	32.9	41.1	48.0	56.6	63.0	64.4	57.9	44.3	34.2	27.4		
		MCDB	21.6	27.3	38.9	50.1	57.6	63.2	69.6	71.3	63.1	49.5	39.8	30.7		
	5%	WB	16.7	18.2	30.5	37.8	45.7	54.5	61.6	62.6	55.7	41.9	31.4	24.7		
		MCDB	18.9	21.1	35.8	45.0	54.4	60.8	68.3	68.4	60.4	47.1	36.4	28.3		
	10%	WB	13.6	13.6	28.1	35.5	43.5	52.9	60.2	61.1	53.4	39.5	29.0	22.5		
		MCDB	15.6	15.9	32.8	41.9	51.0	59.2	66.8	66.6	58.6	44.3	33.3	25.8		
	5% DB	MDBR	14.8	18.1	17.5	14.9	15.6	14.2	13.4	13.1	14.1	13.7	12.6	11.6		
		MCDBR	16.8	20.1	18.2	18.7	21.0	17.8	15.6	15.8	16.3	17.9	16.0	14.4		
		MCWBR	15.0	17.0	13.7	12.4	13.1	10.5	9.1	9.3	11.4	13.1	12.8	12.2		
		MCWBR	16.5	20.0	18.3	18.4	19.0	15.5	14.2	14.1	15.2	17.9	15.4	14.1		
(35) Mean Daily Temperature Range	5% WB	MCWBR	14.9	17.1	13.9	12.7	12.5	11.0	9.8	9.6	11.4	13.8	12.6	12.2		
	Clear-Sky Solar Irradiance	taub	0.260	0.286	0.314	0.358	0.407	0.440	0.426	0.380	0.333	0.304	0.274	0.256		
		taud	2.105	2.096	2.152	2.301	2.212	2.213	2.303	2.413	2.491	2.430	2.171	2.068		
(42) All-Sky Solar Radiation	RadAvg	Ebn at Noon	223	255	274	277	267	258	260	267	269	252	216	196		
		Edh at Noon	22	31	37	36	42	43	38	32	27	23	21	19		
(44) All-Sky Solar Radiation	RadStd		360	703	1203	1560	1762	1892	1751	1487	1075	661	354	245		
			19	29	46	103	121	112	122	62	89	58	24	18		

Historical Trends

		Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47)	Regional (12 neighbors)	N/A	N/A	-3.74	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page